

E- COMMERCE STORE REPORT

Outline

1. Summary of Observation
2. Data cleaning
3. Insight reports

Observations

This is dataset contain records of sales of an online store for the years 2010 and 2011.

In this dataset, we have columns for Customer IDs which uniquely identifies a customer, Invoice No. which marks a transaction made, Unit-price for each item, Quantity of each item purchased,

Description of the Item & Stock-code for each unique item, Date/Time of purchase, Country from which item was purchase was made. I derived the following columns; Revenue generated by each item, Hour of purchase of each item and Month of purchase of item.

Each row represents an item purchase.

I observed that there were of missing Customer-IDs, empty country rows, negative quantity entries. Also, some Invoice No., Stock-code were mixed with letters and duplicate records.

Data cleaning

This involved Finding and replacing values, removing entries with negative quantities, missing Customer-IDs and missing countries.

Insights

This is a detailed insight of the Online sales dataset;

1. Executive summary

The store is performing strongly, with roughly:

- **18,536 items**
- **3,868 orders**
- **4,339 customers**
- **\$8.89M in revenue**
- **Very high item volume** shown on the dashboard, indicating a bulk-heavy retail pattern

The business looks healthy, but it is also **highly concentrated** in a few places: one country United Kingdom, appears to drive most revenue and order volume, a small group of VIP

customers contributes a large share of sales and a few products dominate both quantity sold and revenue.

This is good for short-term performance, but it also creates dependency risk.

2. KPI overview of

Orders and customers

With **18.5k orders** from **4.3k customers**, the store appears to have repeat buying behavior. That suggests the business is not relying only on one-time shoppers. It likely has a base of recurring customers or wholesale-style buyers. We have about 4.3 orders per customer, \$479 revenue per order and \$2,049 revenue per customer. That points to a store with decent customer value and repeat engagement.

Revenue

Total revenue is near **\$8.9M**, which is strong relative to the customer count. This tells us the store is not only selling often, but also generating meaningful value per transaction.

3. Monthly performance insight

The **Monthly Purchase by Revenue & Quantity** chart shows a clear seasonal pattern with

- **January to April:** unstable and softer performance
- **May to August:** moderate recovery and more stable demand
- **September to November:** strong growth surge
- **November:** highest point for both revenue and quantity
- **December:** slight decline from the peak, but still very strong compared to the first half of the year

This looks like a classic retail seasonality curve, weaker start to the year, gradual increment through mid-year, strong holiday or year-end demand in Q4.

Revenue and quantity move almost in the same direction across the months meaning growth is mostly coming from more units sold, demand for product is real, not just pricing noise

4. Hourly sales behavior

The Hourly Purchase by Revenue & Quantity shows very low activity from 6 AM to 8 AM, strongest window from about 10 AM to 1 PM, activity remains decent until around 3 PM and clearly drop after 4 PM which is towards closing time.

This shows that majority of customer traffic comes at late morning to early afternoon.

5. Customer insight

There are two customer ranking charts the **Top 20 customers/VIP by quantity** and **Top 20 customers/VIP by revenue**.

A. One Customer **14646** appears to be the largest buyer by quantity, clearly ahead of the rest.

This may indicate a wholesale or reseller buyer, a corporate buyer or a very loyal high-volume account.

B. Revenue leadership is also concentrated

The top revenue chart shows a few customers well ahead of the rest. The leading Customer **14646** is again near the top, followed by a small group that contributes large revenue.

C. Quantity leaders are not exactly the same as revenue leaders.

6. Product insight

There are also two very useful product views which includes the **Top 10 most purchased products** and **Top 10 products by revenue**. Some products appear to be strong in both volume and value e.g. PAPER CRAFT, LITTLE BIRDIE, REGENCY CAKESTAND 3 TIER, WHITE HANGING HEART T-LIGHT HOLDER etc.

The top quantity list and top revenue list are not identical which means that some items are traffic movers that sell in large numbers but may not bring the most money per unit, while some items are revenue drivers which may sell fewer units but generate more total revenue.

7. Geographic insight

The **United Kingdom**, appears to dominate both order volume and revenue with around **\$7.29M** while total revenue is about **\$8.89M**, making it the leading market that contributes the majority of total sales.

There are secondary visible markets in parts of Western Europe, Australia, North America and a few smaller markets in the Middle East, Africa, and South America

8. Overall business diagnosis

This dashboard suggests the business has **three strong engines**:

1. Seasonal demand engine with Q4 being the strongest period.

2. VIP / repeat buyer which is a relatively small set of customers contributes an outsized value.

Moreover, this is a good base for growth, but it also means the business is concentrated around: a few months, a few customers, a few products and one major country

That concentration is efficient now, but risky if left unmanaged.

9. Risks visible from the dashboard

Customer concentration risk: If top buyers reduce orders, revenue may drop quickly.

Product concentration risk: If a few very valuable products go out of stock, revenue and conversion can fall fast.

Country concentration risk: Heavy dependence on one market makes the business vulnerable to market-specific issues.

Timing concentration risk

A large share of business happens in a narrow time window: months of September – November, peak hours of 10 AM–1 PM. This can stress fulfillment and inventory planning.

10. Recommended actions

Immediate actions

- protect inventory for the top 20 products
- build a VIP watchlist for the top revenue customers
- devise a purchase campaign timing towards the midday buying window
- prepare and secure Q4 stock much earlier.

Medium-term actions

- separate customers into **bulk**, **premium**, and **regular**
- build bundles around top-volume and top-revenue items
- test growth in secondary countries already showing activity
- create reactivation campaigns for customers outside the VIP group